Class18

Yuxuan Jiang (A17324184)

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Background

Pertuissis (whooping cough) is a common lung infection caused by the bacteria B.Pertussis. It can infect anyonebut is most deadly for infants (under 1 year of age).

CDC tracking data

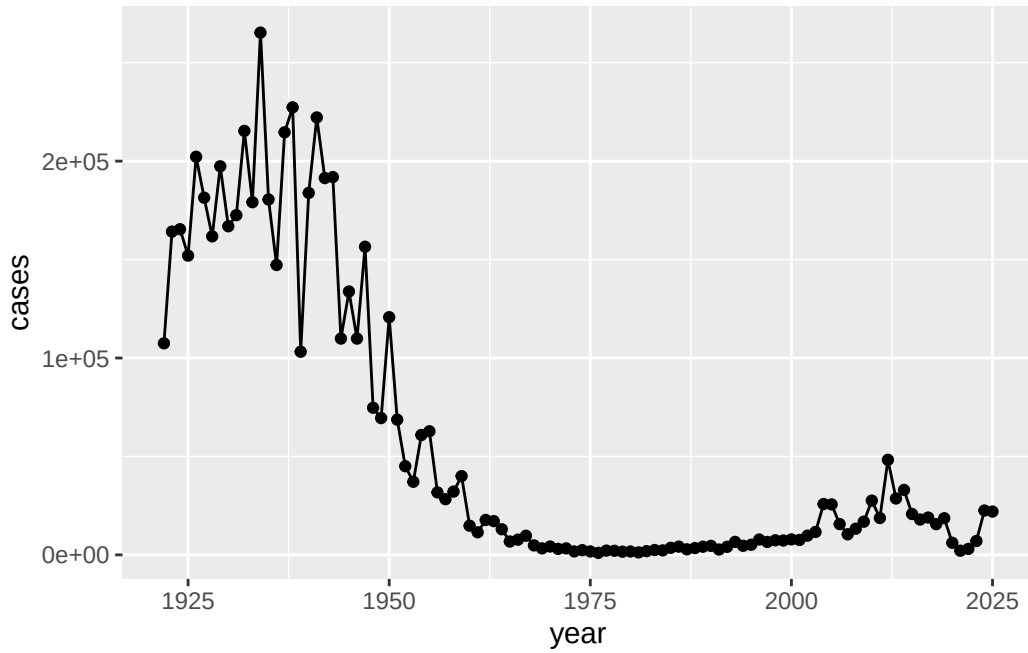
The CDC tracks the number of Pertussis cases:

```
cdc <- data.frame(year = c(1922L,1923L,1924L,1925L,
 1926L,1927L,1928L,1929L,1930L,1931L,
 1932L,1933L,1934L,1935L,1936L,
 1937L,1938L,1939L,1940L,1941L,1942L,
 1943L,1944L,1945L,1946L,1947L,
 1948L,1949L,1950L,1951L,1952L,
 1953L,1954L,1955L,1956L,1957L,1958L,
 1959L,1960L,1961L,1962L,1963L,
 1964L,1965L,1966L,1967L,1968L,1969L,
 1970L,1971L,1972L,1973L,1974L,
 1975L,1976L,1977L,1978L,1979L,1980L,
 1981L,1982L,1983L,1984L,1985L,
 1986L,1987L,1988L,1989L,1990L,
```

```
 1991L,1992L,1993L,1994L,1995L,1996L,
 1997L,1998L,1999L,2000L,2001L,
 2002L,2003L,2004L,2005L,2006L,2007L,
 2008L,2009L,2010L,2011L,2012L,
 2013L,2014L,2015L,2016L,2017L,2018L,
 2019L,2020L,2021L,2022L,2023L,2024L,2025L),
 cases = c(107473,164191,165418,152003,
 202210,181411,161799,197371,
 166914,172559,215343,179135,265269,
 180518,147237,214652,227319,103188,
 183866,222202,191383,191890,109873,
 133792,109860,156517,74715,69479,
 120718,68687,45030,37129,60886,
 62786,31732,28295,32148,40005,
 14809,11468,17749,17135,13005,6799,
 7717,9718,4810,3285,4249,3036,
 3287,1759,2402,1738,1010,2177,2063,
 1623,1730,1248,1895,2463,2276,
 3589,4195,2823,3450,4157,4570,
 2719,4083,6586,4617,5137,7796,6564,
 7405,7298,7867,7580,9771,11647,
 25827,25616,15632,10454,13278,
 16858,27550,18719,48277,28639,32971,
 20762,17972,18975,15609,18617,
 6124,2116,3044,7063,22538,21996)
       )
```

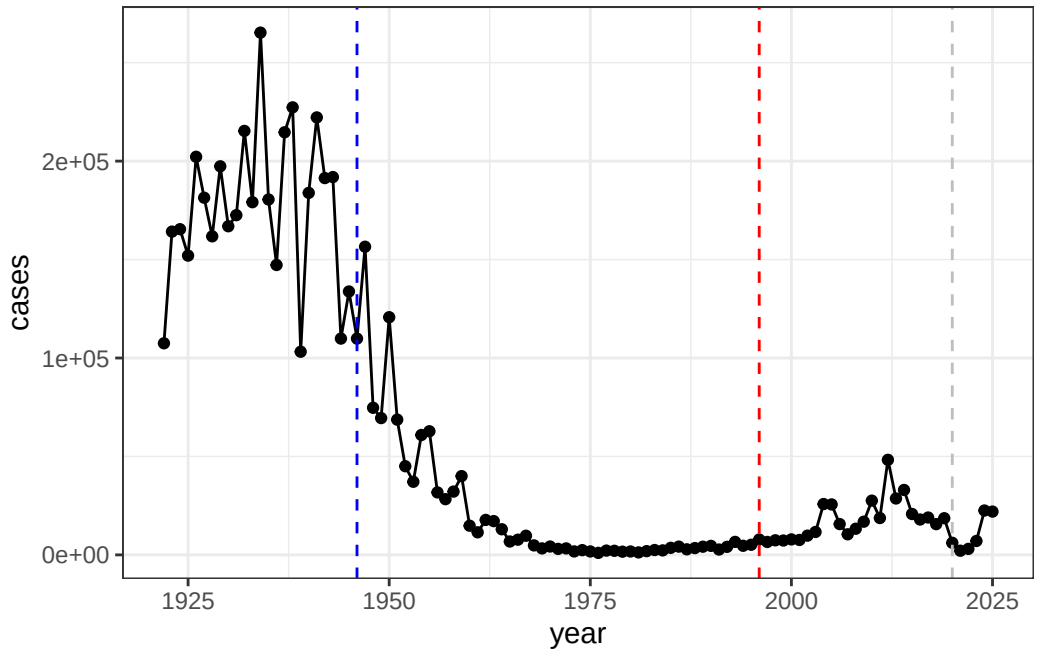
Q1.Year vs cases plot

```
library(ggplot2)
ggplot(cdc)+
  aes(year,cases)+
  geom_point()+
  geom_line()
```



Q2&Q3 Let's add annotation lines for the major milestones of wP vaccination roll-out (1946), the switch to the aP vaccine (1996), COVID-19(2020).

```
library(ggplot2)
ggplot(cdc)+
  aes(year,cases)+
  geom_point()+
  geom_line()+
  theme_bw()+
  geom_vline(xintercept = 1946,col="blue",linetype="dashed")+
  geom_vline(xintercept=1996,col="red",linetype="dashed")+
  geom_vline(xintercept=2020,col="gray",linetype="dashed")
```



The number of the reported cases dropped dramatically after 1946, highly likely to be due to the wP vaccination roll-out. However, the reported cases increased significantly after the roll-out of the newly developed aP vaccine. This may due to the reduced vaccination rate. In addition, the protective effect provided by aP vaccine may fade faster than the wP vaccine. Finally, the development of more sensitive detection techniques may also contribute to the increase of detected cases.

Exploring the CMI-PB data

CMI-PB tracks and makes freely available long-term humoral and cellular immune response data for a large number of individuals who received either DTwP or DTaP combination vaccines in infancy followed by Tdap booster vaccinations. This includes complete API access to longitudinal RNA-Seq, AB Titer, Olink, and live cell assay results directly from their website: https://www.cmi-pb.org/

Basically, make available a large dataset on the immune response to Pertussis. They use a "booster" vaccination as a proxy for pertussis infection.

The CMI-PB API returns JSON data

They make their data available as JSON format API. To read these types of files into R we will use the read_json() function from the jsonlite package.

```r
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q4. How many aP and wP individuals are there in this `subject` table?

```r
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

```r
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc…)? Is this representative of the US population?

```r
table(subject$race,subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

Due to the impact of COVID-19, the sample population is not representative of the US population.

Joining multiple tables

We can now read more tables from the CMI-PB database.

```
library(jsonlite)
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
ab_titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRU
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To make sense of all this data we need to merge all these tables together. Only then will you know that a given Ab measurement (from the `ab_tilter` table) was collected on a certain data (from the `specimen` table) from a certain wP or aP subject (from the `subject` table).

We can use `dplyr` and the `*_join()` family of functions to do this.

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject,specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

Let's do one more `inner_join()` to join the `ab_titer` with all our `meta` data.

```r
library(dplyr)
abdata <- inner_join(ab_titer,meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
```

```
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q. How many different Ab isotype values are in this dataset?

```r
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Q. How many different antigen values are in this dataset?

```r
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA 
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318 
    PD1     PRN      PT     PTM   Total      TT 
   1970    6712    6712    1970     788    6318 
```

> Q12.What are the different $dataset values in abdata and what do you notice about the number of rows for the most "recent" dataset?

```r
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset 
       31520         8085         7301        15050 
```

The most recent dataset has almost double the number of rows compared to the 2021 and 2022 datasets, suggesting that more participants were included in the study in 2023 than in 2021 and 2022.

Let's focus on IgG isotype:

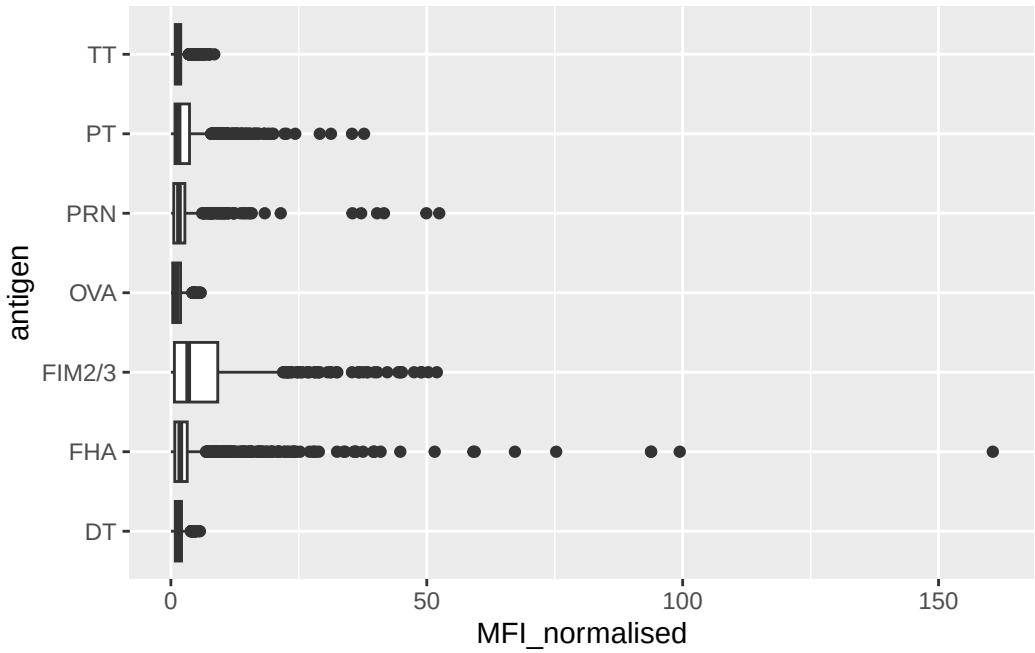
```r
library(dplyr)
igg <- abdata |>
       filter(isotype=="IgG")
head(igg)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
```

```
6 IU/ML                 4.679535          3          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Plot for `MFI_normalised` vs. `antigen`:

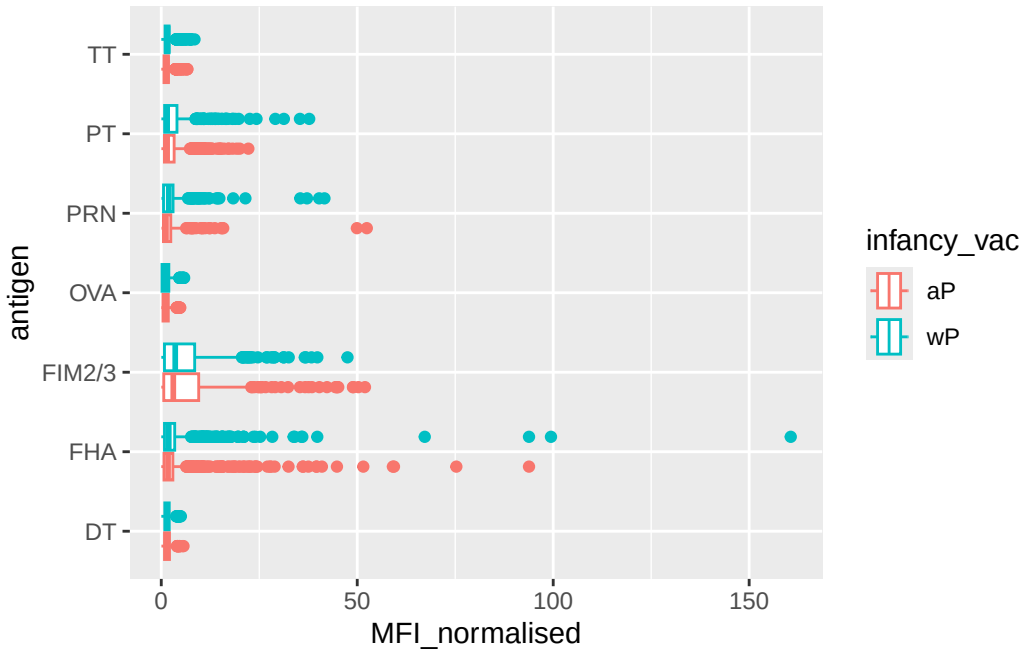
```
library(ggplot2)
ggplot(igg)+
  aes(MFI_normalised,antigen)+
  geom_boxplot()
```



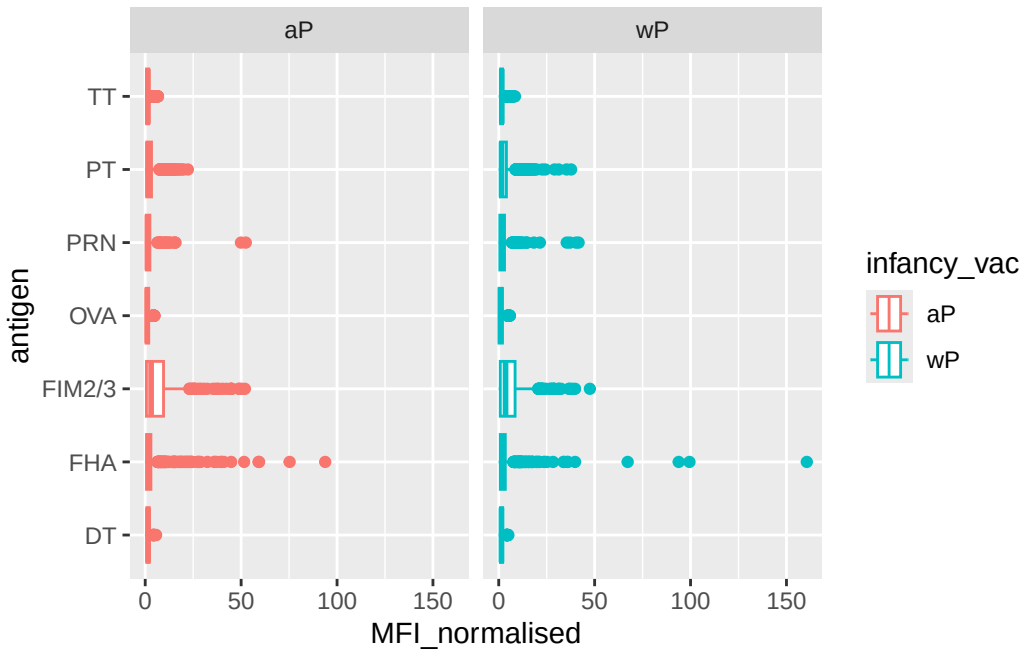
The antigen "PT", "FIM2/3" and "FHA" appears to have the widest range of values.

Q. is there a difference for thses responses between aP and wP individual?

```
library(ggplot2)
ggplot(igg)+
  aes(MFI_normalised,antigen,col=infancy_vac)+
  geom_boxplot()
```

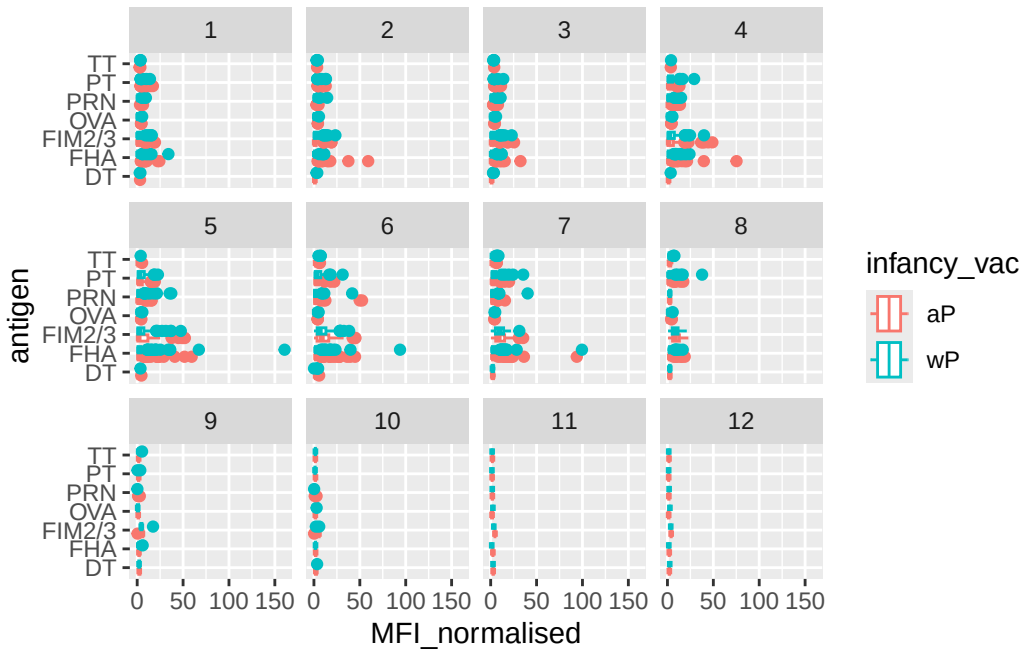



```
library(ggplot2)
ggplot(igg)+
  aes(MFI_normalised,antigen,col=infancy_vac)+
  geom_boxplot()+
  facet_wrap(~infancy_vac)
```



Q.Is there a difference with time (i.e. before booster shot vs. after booster shot)?

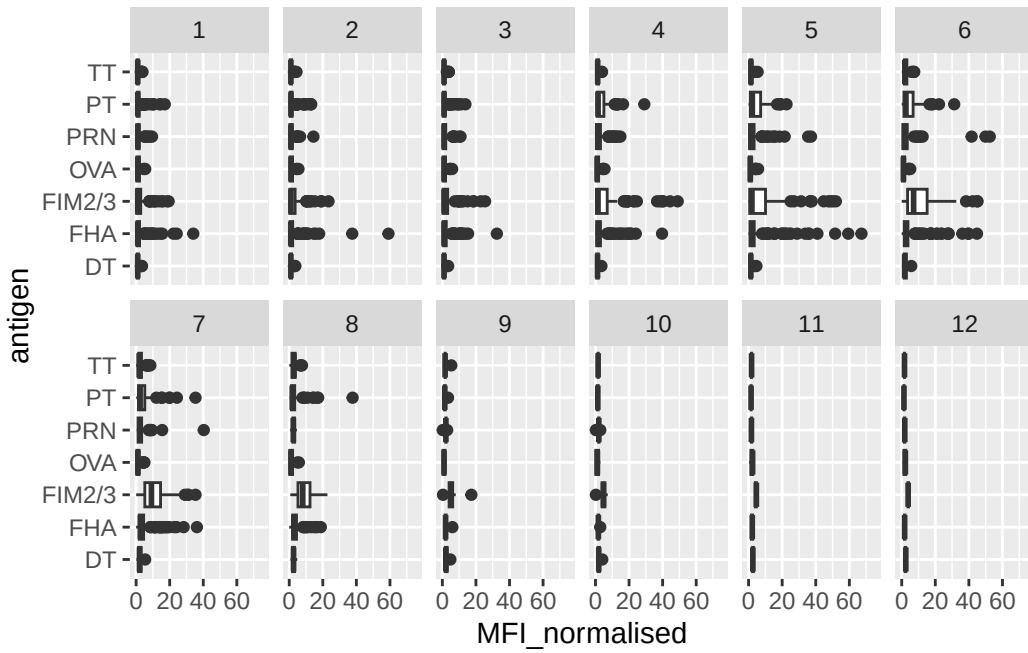
```
library(ggplot2)
ggplot(igg)+
  aes(MFI_normalised,antigen,col=infancy_vac)+
  geom_boxplot()+
  facet_wrap(~visit)
```



Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
    xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

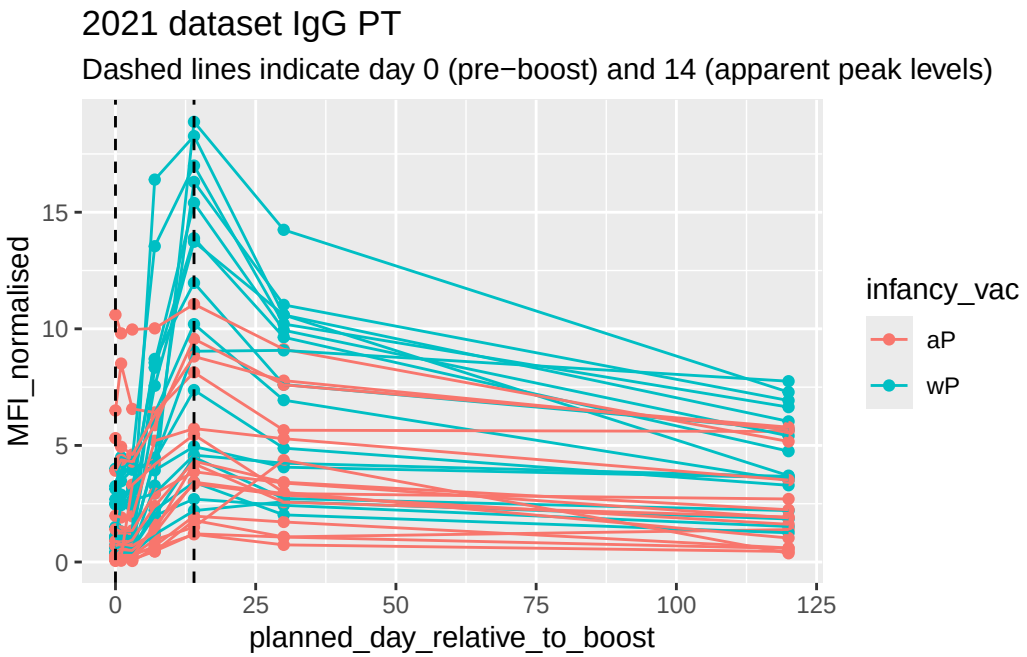
```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

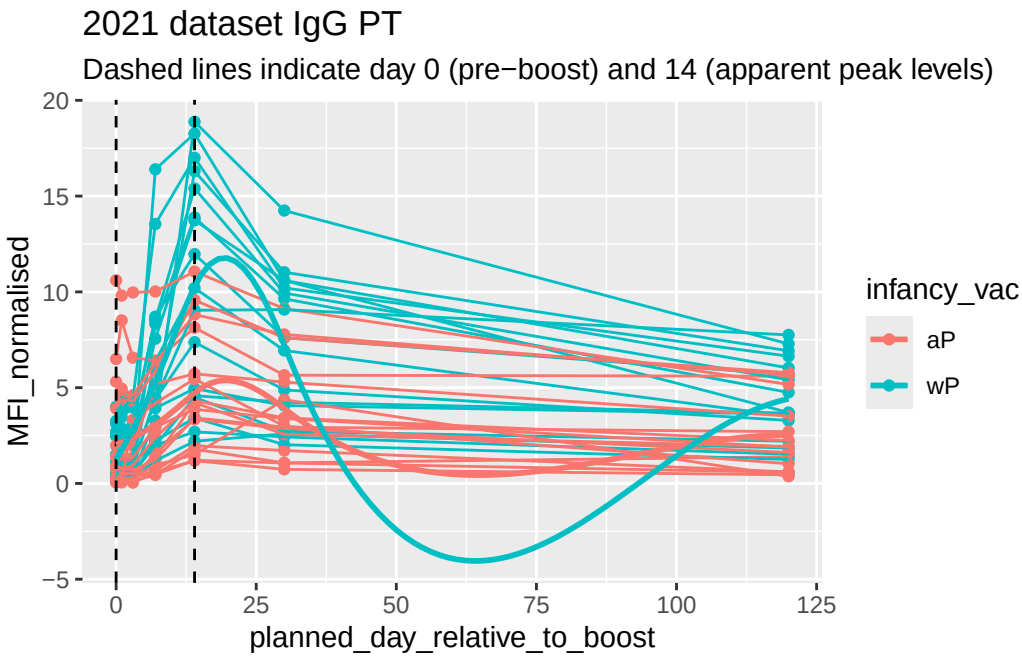
The antigens PT, FIM2/3, and FHA show differences in the level of IgG antibody titers recognizing them over time. Because these antigens are components of the aP vaccine, the observed increase in IgG levels likely reflects the immune activation following vaccination.

```
ab.PT.21 <- abdata |>
             filter(dataset == "2021_dataset",
                    isotype == "IgG",
                    antigen == "PT")
  ggplot(ab.PT.21) +
    aes(planned_day_relative_to_boost,
        MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line()+
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2021 dataset IgG PT",
       subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```



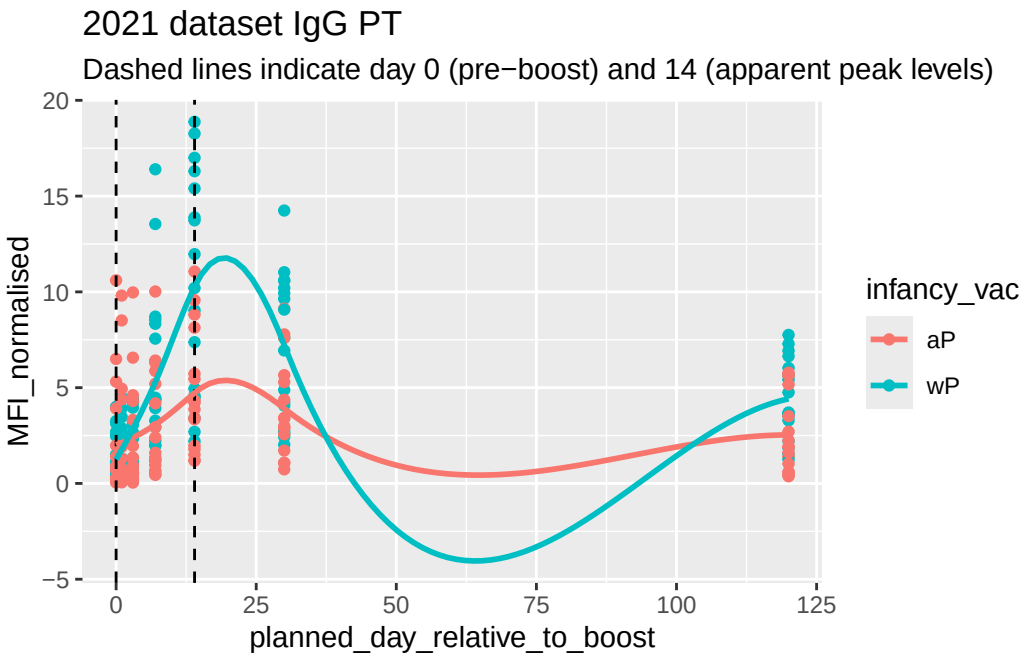

```
ab.PT.21 <- abdata |>
             filter(dataset == "2021_dataset",
                    isotype == "IgG",
                    antigen == "PT")
  ggplot(ab.PT.21) +
    aes(planned_day_relative_to_boost,
        MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line()+
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2021 dataset IgG PT",
       subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")+
    geom_smooth(aes(group = infancy_vac),se=FALSE)
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
ab.PT.21 <- abdata |>
             filter(dataset == "2021_dataset",
                    isotype == "IgG",
                    antigen == "PT")
  ggplot(ab.PT.21) +
    aes(planned_day_relative_to_boost,
        MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2021 dataset IgG PT",
       subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")+
    geom_smooth(aes(group = infancy_vac),se=FALSE)
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```


2021 dataset IgG PT
Dashed lines indicate day 0 (pre−boost) and 14 (apparent peak levels)
MFI_normalised
planned_day_relative_to_boost
infancy_vac
aP
wP
20
15
10
5
0
−5
0
25
50
75
100
125